

Research Methodology in CS (CSEG3060_3)

Lecture 4: Addressing Plagiarism: definition, types (direct, self, mosaic, accidental), and detection tools

Unit 3: Ethical Conduct and Literature Studies • Mapped Course Outcome: CO3

4: Addressing Plagiarism — Definition, Types, and Detection Tools

Course: Research Methodology in CS (CSEG3060_3) **Program:** B.Tech (Computer Science)
Semester: V **Unit:** III — Ethical Conduct and Literature Studies **Course Outcome:** CO3 **Duration:** 60 Minutes

1. Learning Objectives

By the end of this lecture, students will be able to:

1. Define plagiarism in academic and research contexts with precision.
 2. Identify and differentiate between the major types of plagiarism: direct, self, mosaic, and accidental.
 3. Apply recognized citation frameworks (APA, IEEE, MLA) to mitigate attribution errors.
 4. Operate commonly used plagiarism detection tools and interpret similarity reports critically.
 5. Recognize the ethical, legal, and professional consequences of plagiarism in computer science research.
 6. Formulate institutional and personal strategies to uphold academic integrity.
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2. Introduction

The proliferation of digital repositories, open-access journals, generative AI tools, and code-sharing platforms has fundamentally transformed how knowledge is produced, disseminated, and consumed in computer science. While these advances accelerate innovation, they simultaneously expand the surface area for academic misconduct. **Plagiarism** remains one of the most frequently reported violations in higher education and among the leading causes of paper retractions globally.

A 2023 analysis by *Retraction Watch* indicated that plagiarism-related retractions in computer science rose by approximately 38 percent over the previous decade, with a significant proportion

attributed to inadequate paraphrasing and improper reuse of prior work. For computer scientists, whose research outputs often include source code, algorithms, datasets, and documentation, the boundaries of acceptable reuse require careful definition.

This lecture provides a rigorous examination of plagiarism, its typology, detection mechanisms, and the practical strategies researchers must adopt to maintain ethical integrity.

3. Defining Plagiarism

3.1 Etymology and Conceptual Foundation

The term *plagiarism* derives from the Latin word **plagiarius**, meaning "kidnapper," reflecting its historical association with theft of intellectual property. In academic literature, plagiarism is generally defined as the appropriation of another person's ideas, processes, results, or words without giving appropriate credit.

3.2 Formal Definitions

Several authoritative bodies have provided formal definitions:

Source	Definition
Office of Research Integrity (ORI), USA	"Plagiarism is the appropriation of another person's ideas, processes, results, or words without giving appropriate credit."
Council of Science Editors (CSE)	Plagiarism involves using the work, ideas, or words of others without proper attribution, irrespective of intent.
IEEE Publication Policy	Plagiarism includes "copying ideas, text, data, software, or any other material without proper citation."
UGC (India) Regulations 2018	Plagiarism is the practice of taking someone else's work or ideas and passing them off as one's own.

3.3 Core Constituents of Plagiarism

For an act to constitute plagiarism, the following elements must generally be present:

- 1. Originality of the source:** The appropriated material must originate from a verifiable source.
- 2. Lack of attribution:** The borrower fails to provide credit commensurate with the borrowed content.
- 3. Intent or negligence:** Either deliberate deception or reckless disregard for citation norms.

4. **Scope of appropriation:** The extent may range from a phrase to an entire manuscript.

4. Typology of Plagiarism

Plagiarism manifests in multiple forms, each with distinct characteristics and remediation strategies. The taxonomy below synthesizes classifications from the **ORI**, **Turnitin**, **iThenticate**, and contemporary research ethics literature.

4.1 Direct Plagiarism

Definition: Direct plagiarism involves the verbatim copying of text, code, data, or visual content from a source without quotation marks, citation, or any indication of the original author.

Characteristics:

- Identical reproduction of sentences or paragraphs.
- No use of quotation marks.
- Complete absence of citations.
- Often the most blatant and easily detected form.

Example:

Suppose a student submits the following as part of their thesis on machine learning:

"Deep learning is a subset of machine learning that employs neural networks with many layers. These networks can learn complex patterns in large datasets, enabling breakthroughs in image recognition, natural language processing, and autonomous systems."

If this paragraph is taken word-for-word from a published article without quotation marks or citation, the act constitutes **direct plagiarism**.

Severity: Highest, particularly when the copied material forms a substantial portion of the submission.

Illustration of textual overlap (conceptual representation):

$$\text{Overlap Ratio} = \frac{\text{Matched Words or Phrases}}{\text{Total Words or Phrases in Document}} \times 100\%$$

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A direct copy of a paragraph typically yields an overlap ratio exceeding **85–90 percent** within that segment.

4.2 Self-Plagiarism (Auto-Plagiarism)

Definition: Self-plagiarism occurs when an author reuses their own previously published work, or portions thereof, without proper disclosure or citation. Although the author retains ownership of the ideas, academic publishing ethics require transparency about prior dissemination.

Manifestations in CS Research:

- Submitting the same research paper to multiple journals (duplicate publication).
- Reusing substantial portions of methodology sections across papers without citation.
- Recycling previously published datasets, code, or algorithms in new publications without acknowledgment.

Why It Is Problematic:

1. **Misrepresentation of novelty:** Reviewers and readers may believe the work is original.
2. **Copyright infringement:** Publishers often hold exclusive rights to published content.
3. **Violation of peer review:** Multiple submissions waste reviewer effort and editorial resources.

Mitigation:

- Disclose prior related work in the cover letter and manuscript.
- Use quotation marks and citations when reusing exact text.
- Obtain permission from the original publisher when required.
- Limit textual overlap across manuscripts to methodology citations and standard references.

Example: A researcher publishes a paper in *Journal A* in 2022 on "Efficient Graph Neural Networks for Social Networks." In 2024, they submit a substantially similar paper to *Conference B* without citing the 2022 publication. This constitutes **self-plagiarism**, even though the author owns both works.

4.3 Mosaic Plagiarism (Patchwork Plagiarism)

Definition: Mosaic plagiarism, also known as patchwork plagiarism, involves combining text, ideas, or data from multiple sources while making superficial modifications such as synonym substitution, sentence restructuring, or paraphrasing, without proper citation.

Characteristics:

- No single source contributes a dominant share; the document is a "mosaic."
- Modifications are superficial and do not reflect genuine understanding.
- Citations are either absent or selectively applied.
- Detection requires sophisticated semantic analysis rather than simple string matching.

Example:

Original source:

"Convolutional neural networks have revolutionized image classification by automatically learning hierarchical features from raw pixel data."

Mosaic version (without citation):

"CNNs have transformed image categorization by autonomously extracting multi-level representations directly from unprocessed pixel inputs."

Despite the change in word choice, the structure, idea, and intellectual contribution remain substantially unchanged. This is **mosaic plagiarism**.

Detection Challenge: Mosaic plagiarism is particularly resistant to detection by exact-string matching algorithms. Modern detection tools employ **semantic similarity models** based on natural language processing to identify such cases.

The similarity between original and paraphrased text can be quantified using cosine similarity over embedding vectors:

$$\text{Similarity}(u, v) = \frac{u \cdot v}{\|u\| \|v\|} \quad \text{Similarity}(u, v) = \frac{u \cdot v}{\|u\| \|v\|}$$

where u and v are vector representations of the original and submitted passages, respectively. A cosine similarity above approximately **0.80** typically signals significant semantic overlap.

4.4 Accidental Plagiarism

Definition: Accidental plagiarism arises when plagiarism occurs without intent to deceive. Common causes include ignorance of citation conventions, careless note-taking, or structural imitation of a source.

Common Causes:

1. **Forgotten citations:** Reproducing information from a source without recording the reference.
2. **Improper paraphrasing:** Failing to sufficiently transform the original wording while neglecting quotation marks.
3. **Unintentional structural replication:** Following the organizational outline of a source too closely.
4. **Cryptomnesia:** Subconscious reproduction of material encountered previously but believed to be original.
5. **Misattribution:** Citing a secondary source rather than the original, or vice versa.

Example: A student summarizes a paragraph from a Wikipedia article, omits the citation because they "thought it was common knowledge," and submits the text in an assignment. Although unintentional, this constitutes **accidental plagiarism** under most institutional policies.

Important Note: The absence of malicious intent does not absolve the author. Most academic integrity policies treat accidental plagiarism as a punishable offense, albeit typically with lesser sanctions on the first offense.

4.5 Additional Recognized Forms

For completeness, the following types are also recognized in contemporary scholarship:

Type	Description
Idea Plagiarism	Appropriating concepts or theoretical frameworks without attribution.
Code Plagiarism	Copying source code, algorithms, or software components without credit. Particularly relevant in CS.
Data Plagiarism	Reusing datasets without permission or acknowledgment.
Image Plagiarism	Reproducing figures, diagrams, or screenshots without citation.
Translation Plagiarism	Translating a foreign-language source and presenting it as original.
AI-Assisted Plagiarism	Submitting AI-generated text as one's own, or paraphrasing AI outputs without disclosure.

5. Plagiarism Detection Tools

Plagiarism detection has evolved from simple string-matching algorithms to sophisticated systems incorporating natural language processing, machine learning, and large-scale database

comparison.

5.1 Classification of Detection Approaches

Approach	Methodology	Strengths	Limitations
String Matching	Exact or near-exact text comparison	Fast, transparent	Fails on paraphrased content
Fingerprinting	Hash-based indexing of n-grams	Efficient for large corpora	Sensitive to word order changes
Stylometry	Statistical analysis of writing style	Detects ghostwriting	Requires substantial reference text
Semantic Similarity	NLP-based embedding comparison	Identifies paraphrasing	Computationally intensive
Citation Analysis	Examination of citation patterns	Detects missing citations	Limited to structured documents

5.2 Prominent Detection Tools

5.2.1 Turnitin

- **Use Case:** Widely deployed in academic institutions for student submissions.
- **Database:** Compares against internet sources, student paper repositories, and publications.
- **Output:** Similarity index (percentage) and highlighted matching passages.
- **Strengths:** Comprehensive web index, integration with learning management systems.
- **Limitations:** Does not detect well-paraphrased text or ideas; thresholds vary by institution.

5.2.2 iThenticate

- **Use Case:** Preferred tool for journal publishers and research manuscripts.
- **Database:** CrossCheck database of published scholarly content.
- **Strengths:** High-quality scholarly comparison; integrated with editorial workflows.
- **Limitations:** Subscription required; primarily text-based.

5.2.3 Grammarly Plagiarism Checker

- **Use Case:** Lightweight checking integrated with writing assistance.

- **Strengths:** Convenient for early-stage drafts; provides writing feedback.
- **Limitations:** Limited database compared to Turnitin.

5.2.4 Copyscape

- **Use Case:** Web-based plagiarism detection for online content.
- **Strengths:** Effective for detecting unauthorized online republication.
- **Limitations:** Less suitable for academic manuscript review.

5.2.5 PlagScan

- **Use Case:** Enterprise and institutional deployments.
- **Strengths:** Configurable sensitivity; multi-format support.
- **Limitations:** Smaller database than Turnitin.

5.2.6 MOSS (Measure of Software Similarity)

- **Use Case:** Specifically designed for **source code** plagiarism detection.
- **Strengths:** Handles structural similarity in programming assignments.
- **Methodology:** Uses winnowing and fingerprinting algorithms adapted for code.
- **Relevance:** Particularly important for CS courses and programming assessments.

5.2.7 JPlag

- **Use Case:** Open-source plagiarism detection for code, supporting multiple programming languages.
- **Strengths:** Language-agnostic token-based comparison.
- **Relevance:** Used in academic settings for algorithm-based detection.

5.2.8 Urkund (Ouriginal)

- **Use Case:** European institutional deployments.
- **Strengths:** Multilingual support; integration with LMS platforms.

5.2.9 AI-Generated Text Detectors

With the emergence of large language models (LLMs), new tools have emerged:

- **GPTZero**

- **Originality.ai**
- **Turnitin AI Writing Detection**
- **Copyleaks AI Content Detector**

These tools analyze perplexity, burstiness, and stylistic patterns to estimate the likelihood of AI-generated content. However, their reliability remains contested, with documented false-positive rates.

5.3 Interpreting Similarity Reports

A high similarity index does not automatically imply plagiarism. Proper interpretation requires:

1. **Source identification:** Determine whether matches originate from properly cited sources.
2. **Quoted material recognition:** Exclude legitimately quoted passages with quotation marks.
3. **Common knowledge exclusion:** Phrases or terminology widely accepted as common knowledge.
4. **Methodology overlap:** Standard methods or equations may appear across multiple papers.
5. **Reference list exclusions:** Most tools exclude bibliography sections; verify this is configured correctly.

Recommended Thresholds (indicative, not universal):

Similarity Index	Typical Interpretation
< 10%	Generally acceptable
10–20%	Review required
20–40%	Significant overlap; likely requires revision
> 40%	High concern; requires major restructuring or proper citation

Note: These thresholds are institution- and journal-specific. For instance, IEEE permits up to 30 percent similarity for review articles under certain conditions.

6. Case Studies

6.1 Case Study 1: The Hwang Woo-suk Affair (Scientific Misconduct)

Background: In 2004–2005, South Korean researcher Hwang Woo-suk published landmark papers in *Science* claiming successful cloning of human embryonic stem cells. Investigations

revealed that data had been fabricated and key figures duplicated from earlier publications.

Plagiarism Element: Although primarily a data fabrication case, the incident included image reuse and uncredited appropriation of research methodologies.

Outcome: Papers retracted, funding revoked, criminal conviction, and subsequent reforms in research integrity oversight.

Lesson: Plagiarism detection tools, when applied rigorously to figures and data, can reveal inconsistencies that textual review misses.

6.2 Case Study 2: Retraction of a Computer Science Paper for Self-Plagiarism

Background: A 2019 paper in a Springer journal was retracted after editors identified that the authors had published substantially overlapping content in two separate venues without cross-citation.

Lesson: Self-plagiarism is a serious ethical violation, even when the author owns both works. Researchers must maintain a careful publication record and disclose prior related work.

6.3 Case Study 3: Code Plagiarism in Programming Courses

Background: Universities have reported widespread code plagiarism in introductory programming courses. Studies using MOSS have shown that in some cohorts, **15–25 percent** of submitted programming assignments contain significant code overlap with other submissions.

Mitigation Strategies:

- Use MOSS or JPlag for automated detection.
- Design assignments with unique parameters per student.
- Conduct viva voce examinations to verify understanding.
- Educate students about academic integrity policies from the outset of the course.

7. Consequences of Plagiarism

7.1 Academic Consequences

- Failure or zero grade in the submitted work.
- Course failure or academic probation.
- Suspension or expulsion from the program.
- Revocation of degrees in severe cases.

7.2 Professional Consequences

- Retraction of published papers.
- Blacklisting from journals and conferences.
- Loss of research funding.
- Termination of employment.

7.3 Legal Consequences

- Copyright infringement lawsuits.
- Breach of publisher agreements.
- Damages and injunctions in civil proceedings.

7.4 Reputational Consequences

- Long-term damage to professional credibility.
- Difficulty in securing future collaborations.
- Public retraction notices affecting future publications.

8. Best Practices to Avoid Plagiarism

1. **Begin research with a structured literature log:** Maintain detailed records of sources, including quotations and paraphrases.
2. **Use reference management tools:** Employ Zotero, Mendeley, or EndNote to organize citations systematically.
3. **Master citation styles:** Familiarize yourself with the citation style mandated by your target venue (APA, IEEE, ACM, MLA, Chicago).
4. **Paraphrase responsibly:** True paraphrasing requires comprehending the original, restructuring the argument, and citing the source.
5. **Use quotation marks** for any verbatim text exceeding standard terminology.
6. **Attribute ideas, not just text:** Even paraphrased concepts require citation.
7. **Disclose prior work:** When extending your own publications, transparently reference them.
8. **For code, attribute origins:** Open-source licenses (MIT, GPL, Apache) require specific attribution formats.
9. **Use AI tools responsibly:** Disclose AI assistance when required by the venue; verify AI-generated content against sources.

10. **Run pre-submission checks:** Use institutional access to Turnitin or iThenticate before submitting.

9. Summary

- **Plagiarism** is the uncredited appropriation of another's intellectual work, encompassing ideas, text, data, and code.
 - The principal types include **direct, self, mosaic, and accidental** plagiarism, each with distinct characteristics and severities.
 - Modern detection tools employ **fingerprinting, NLP-based semantic similarity, and stylometric analysis** to identify overlaps.
 - Similarity indices must be interpreted contextually; quoted material, references, and common knowledge affect interpretation.
 - Computer science research introduces unique challenges, particularly regarding **code plagiarism and AI-assisted authorship**.
 - Adopting rigorous citation practices, using reference management tools, and conducting pre-submission checks are essential safeguards.
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10. Exam Preparation Tips and Practice Questions

10.1 Preparation Tips

1. **Distinguish definitional nuances:** Be precise about the differences between direct, self, mosaic, and accidental plagiarism. Examiners often test comprehension of boundaries.
2. **Memorize institutional definitions:** Familiarize yourself with UGC, IEEE, and ACM definitions.
3. **Understand detection algorithms:** Be prepared to explain fingerprinting, n-gram analysis, and cosine similarity qualitatively.
4. **Know the tools:** Associate each tool with its primary use case (e.g., MOSS for code, iThenticate for manuscripts).
5. **Practice applied scenarios:** Expect case-study questions where you must identify the type of plagiarism and recommend remediation.
6. **Cite correctly in answers:** Use proper citation style even in exam answers if referencing course materials or specific sources.

10.2 Practice Questions

Short Answer Questions (2–3 marks each)

1. Define plagiarism as per the Office of Research Integrity (ORI).
2. Differentiate between direct plagiarism and mosaic plagiarism with one example each.
3. What is self-plagiarism? Why is it considered unethical even though the author owns the original work?
4. List any four causes of accidental plagiarism.
5. Name two plagiarism detection tools specifically designed for source code.

Long Answer Questions (5–10 marks each)

1. Discuss the four major types of plagiarism in detail. Provide a real-world or hypothetical example for each.
2. Explain the working principles of plagiarism detection tools. Compare string-matching approaches with semantic similarity methods.
3. Describe the consequences of plagiarism at academic, professional, and legal levels. What policies has UGC (India) framed to address plagiarism in higher education?
4. MOSS and JPlag are used for source code plagiarism detection. Compare their methodologies and explain why specialized tools are needed for code.
5. With the rise of generative AI tools such as ChatGPT, how has the definition and detection of plagiarism evolved? Discuss the ethical responsibilities of researchers.

Application-Based Question

You are reviewing a manuscript in which the authors have reused four paragraphs verbatim from their own 2021 conference paper without citation or quotation marks, and have paraphrased two additional paragraphs from a 2019 journal article without attribution. Identify each instance of plagiarism, classify its type, and recommend the corrective action in each case.

11. Recommended Further Reading

1. **Roig, M. (2015).** *Avoiding Plagiarism, Self-Plagiarism, and Other Questionable Writing Practices: A Guide to Ethical Writing*. Office of Research Integrity.
2. **Anderson, M. S., & Steneck, N. H. (2011).** *The Problem of Plagiarism*. Urologic Oncology.
3. **Wager, E. (2014).** Defining and responding to plagiarism. *Learned Publishing*, 27(1), 33–42.
4. **IEEE Publication Services and Products Board Operations Manual** — Sections on author ethics and plagiarism.
5. **UGC (India) (2018).** *University Grants Commission (Promotion of Academic Integrity and Prevention of Plagiarism in Higher Educational Institutions) Regulations, 2018*.

End of Lecture 4 Notes.